Scope-aware positioning against modern GPU simulation

The project occupies an unusual intersection; headline errors from other papers are not pooled with our launch-level MAPE.

System / model family	native SASS	event / causal	sectored L1/L2	NCU post-join	full real corpus	typed unknowns
GPGPU-Sim execution-driven baseline	—	~	✓	—	~	—
Accel-Sim validated SASS framework	✓	~	✓	✓	~	—
MGPUSim / Akita event/protocol model	—	✓	✓	—	~	~
NVArchSim industrial event-driven context	~	✓	~	~	—	~
Modern GPU core MICRO 2025 core fidelity	✓	~	~	✓	~	~
Sim-FA / GTSim async / tile abstraction	~	✓	~	✓	~	~
PipeWeave / TileSight analytical / learned / tile	—	~	~	✓	✓	~
Our replay stack trace-to-frontier contract	✓	✓	✓	✓	✓	✓

Reading rule ✓ = in-scope evidence; ~ = partial/different abstraction; — = not claimed. The matrix compares capability intersections, not a single quality score.

Caption. The strongest positioning is not a generic ‘faster simulator’ claim. It is the intersection of native trace identity, causal sector replay, explicit frontier semantics, independent NCU joins, and complete 286/339-root evidence.

Sources: Accel-Sim, MGPUSim/Akita, NVArchSim, MICRO 2025 modern-core model, Sim-FA, GPU-Tile-Sim/GTSim, PipeWeave, TileSight, and local SOTA audit.